

Supplementary Material

Stable Landmark Identity and Unlabeled Adaptation: A ConvNeXt Framework for FoundUS 2026

1 Controlled Adaptation Protocol

The controlled adaptation study used the same sequence grouped folds, labeled image schedule, residual adapter architecture, optimization settings, and evaluation protocol for the labeled donor control and unlabeled donor adaptation. In both conditions, the content images retained their manual landmark targets, while the donor images supplied appearance statistics. Donor source was the comparison variable. Evaluation used epoch 8 and residual scale 1.0 for every fold. The pooled held-out set contained 6,700 images and 27,472 landmark points.

The unlabeled donor pool contained 182,870 readable images after exact SHA256 deduplication. Each canonical unlabeled image contributed once per fold to the donor schedule. Fetal femur had no corresponding unlabeled pool and followed the supervised branch in both controlled conditions.

The hard pseudo-label baseline used the same ConvNeXt predictor, residual adapter architecture, fold splits, labeled schedule, and unlabeled coverage. Predictions of the fixed supervised reference on unlabeled images were decoded in the internal channel representation and re-encoded as Gaussian heatmaps for adapter supervision.

2 Fold-wise Results and Sensitivity

Table 1. Fold-wise matched donor comparison. Deltas denote unlabeled donor adaptation minus labeled donor control.

Fold	Δ MRE	Δ MAE	Δ Proxy
0	-0.0409	-0.0303	-0.0356
1	-0.0854	-0.0154	-0.0504
2	-0.0460	-0.0118	-0.0289
3	-0.0076	-0.0031	-0.0053
4	-0.0006	-0.0167	-0.0086
Mean	-0.0361	-0.0154	-0.0258
SD	0.0340	0.0099	0.0189

The combined proxy improved in all five folds. The fold-wise values are reported descriptively; no statistical significance claim is made.

The response to residual scale was non-monotonic. Unit scale was retained for the primary comparison because it had been fixed before the evaluation over five folds.

Table 2. Fold-0 sensitivity diagnostics. Lower proxy is better. The default reference-consistency weight was $\lambda_a = 0.25$. These diagnostics were not used for checkpoint or hyperparameter selection in the primary comparison over five folds.

Setting	MRE	Measurement MAE	Proxy
Weak appearance range [0.10, 0.30]	24.5731	17.8328	21.2029
Default appearance range [0.25, 0.60]	24.5191	17.8146	21.1668
Strong appearance range [0.50, 0.80]	24.5533	17.8069	21.1801
No reference consistency ($\lambda_a = 0$)	24.5909	17.8563	21.2236

Table 3. Residual-scale diagnostic on fold 0 for unlabeled donor adaptation.

Residual scale	Combined proxy
0.00	21.3623
0.25	21.2139
0.50	21.0588
0.75	21.1496
1.00	21.1668

The hard pseudo-label baseline achieved pooled MRE 23.1593, measurement MAE 16.9242, and combined proxy 20.0417. Unlabeled donor adaptation achieved 23.0815, 16.9964, and 20.0390, respectively. The two approaches therefore exhibited different localization and measurement behavior, with a slightly lower combined proxy for appearance adaptation.

3 Computational Cost

Table 4. Additional adaptation cost under the controlled protocols. Wall-clock values are reported only where comparable records were available.

Condition	Steps/fold	Parameters	Unlabeled images	Time/fold
Supervised reference	0 additional	0 additional	0	—
Labeled donor control	12,552–12,568	899,136	0	—
Unlabeled donor adaptation	12,552–12,568	899,136	182,870	21.0–25.1 min
Hard pseudo-label baseline	12,552–12,568	899,136	182,870	32.8–34.0 min

Recorded wall-clock values were obtained on an RTX 5090. The labeled and unlabeled donor conditions used the same optimization step counts and trainable adapter parameter count. A complete comparable wall-clock record was not available for every historical labeled donor run, so no wall-clock difference between the two donor conditions is inferred.

4 Landmark Identity Audit

The identity audit used fixed held-out predictions from the supervised reference. It involved no retraining, checkpoint reselection, or modification of the submit-

ted predictions. A within-pair swap was evaluated only as an oracle analysis of endpoint correspondence.

Table 5. Summary of the endpoint correspondence audit. Oracle gain is the reduction in MRE when the lower-error correspondence is selected for analysis.

Group	Pair instances	Flip rate	Oracle MRE gain
A4C horizontal	368	32.07%	29.9871
A4C vertical	368	0.27%	0.0127
PSAX	86	20.93%	5.4384
Other-task controls	4,914	4.46%	0.2690

Across all 5,736 audited endpoint pairs, swapping endpoint correspondence left the derived measurement value and measurement MAE unchanged. The audit therefore characterizes landmark correspondence independently of the derived distance. It is used as evidence for the representation problem and does not estimate a causal performance gain from canonicalization.

5 Final Distillation Details and Task Results

Five fixed fold models generated mean soft heatmap targets for the complete readable official unlabeled pool. Coordinate dispersion across teachers defined a continuous reliability weight for each landmark, with a fixed floor of 0.10 and a task relative scale based on the 75th percentile of teacher dispersion.

The final student used a fixed ConvNeXt predictor, fixed base parameters of the ultrasound foundation model, and a fixed fusion module. For the eight tasks with unlabeled data, task specific rank-4 Low-Rank Adaptation (LoRA) modules were applied to the final four Transformer blocks of the foundation encoder, and a task specific contextual correction head refined the reference heatmaps. Only the active task LoRA parameters and correction head were optimized. The fetal femur route, which had no official unlabeled data, remained unchanged during the distillation stage.

Training used Kullback–Leibler (KL) divergence and decoded coordinate consistency weighted by reliability. The coordinate term had weight 0.05, with correction and LoRA regularization weights of 10^{-5} each. The complete readable unlabeled pool contained 182,870 images.

The task-level results show heterogeneous changes across measurements. Aggregate official validation MRE and MAE were 24.7986 and 25.9233 for the final model. On the hidden test set, the same final model achieved Average MRE 24.9356 and Average MAE 26.8758.

Table 6. Official validation results by task for the supervised ConvNeXt ensemble and final distilled model. Deltas denote final model minus supervised ConvNeXt ensemble.

Task	MRE			MAE		
	Reference	Final	Δ	Reference	Final	Δ
A4C	28.965	29.693	+0.728	16.071	17.524	+1.453
AOP	14.886	15.103	+0.217	10.556	11.098	+0.542
FA	18.578	17.892	-0.686	81.497	75.713	-5.784
FUGC	9.167	9.252	+0.085	8.139	8.458	+0.319
HC	45.210	45.834	+0.624	69.191	71.339	+2.148
IVC	34.852	25.329	-9.523	11.843	5.873	-5.970
PLAX	15.315	15.810	+0.495	8.656	8.958	+0.302
PSAX	41.627	39.887	-1.740	16.593	18.638	+2.045
Fetal femur	22.209	24.387	+2.178	18.596	15.709	-2.887